

# ORF309 Midterm 2

## Comprehensive Study Guide

Analysis of 7 Past Exams  
Topics, Frequencies, and Learning Strategy

Created: March 31, 2026

**Exam Coverage:** Slides 7–23 (Post-Midterm 1)

**Exams Analyzed:** F2025, S2025, F2022, S2022, 2021, 2019, 2017

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# 1 Executive Summary

## Key Insight

**Bottom Line:** This exam is highly predictable! Three topics appear in 100% of exams and account for  $\sim 75\%$  of the total points. Master these three techniques and you'll be well-prepared.

## 1.1 Critical Statistics

Based on analysis of 7 past Midterm 2 exams:

- **Slides Covered:** 7–23 (everything after Midterm 1)
- **Exam Format:** 3 questions, 30 points each, 90 total points
- **Time:** 105 minutes for problems + 15 minutes for submission
- **Most Predictable Topics:** Poisson Processes, Random Walks, Joint Densities

## 1.2 The Big Three Topics

1. **Poisson Processes** (Slides 16–19): Appears in **7/7 exams (100%)**
  - Average weight:  $\sim 40\%$  of exam
  - Key techniques: Thinning, superposition, competing processes, conditional expectations
2. **Random Walks & First-Step Analysis** (Slides 20–22): Appears in **7/7 exams (100%)**
  - Average weight:  $\sim 35\%$  of exam
  - Key techniques: Hitting probabilities, expected hitting times, gambler's ruin
3. **Joint & Conditional Densities** (Slides 11–13): Appears in **6/7 exams (86%)**
  - Average weight:  $\sim 25\%$  of exam
  - Key techniques: Building joint PDFs, conditional expectations, law of total expectation

## 1.3 Also Important

- **Exponential Distribution & Lifetimes** (Slides 11, 14–15): 5/7 exams (71%)
- **Brownian Motion** (Slide 23): 3/7 exams (43%), increasingly common in recent exams

## 2 Topic Frequency Analysis

Table 1: Topic Appearance Frequency Across 7 Past Exams

| Topic                       | Frequency  | Slides    | Priority    |
|-----------------------------|------------|-----------|-------------|
| Poisson Processes           | 7/7 (100%) | 16–19     | CRITICAL    |
| Random Walks                | 7/7 (100%) | 20–22     | CRITICAL    |
| Joint/Conditional Densities | 6/7 (86%)  | 11–13     | CRITICAL    |
| Law of Total Expectation    | 6/7 (86%)  | 11–12     | CRITICAL    |
| Thinning Property           | 6/7 (86%)  | 17        | CRITICAL    |
| First-Step Analysis         | 7/7 (100%) | 20–21     | CRITICAL    |
| Exponential/Lifetimes       | 5/7 (71%)  | 11, 14–15 | HIGH        |
| Competing Processes         | 5/7 (71%)  | 16–17     | HIGH        |
| Biased Random Walks         | 4/7 (57%)  | 22        | HIGH        |
| Brownian Motion             | 3/7 (43%)  | 23        | MEDIUM-HIGH |

## 3 Critical Priority Topics

### CRITICAL PRIORITY

Learn these topics FIRST. They appear in 80%+ of exams and account for ~75% of exam content.

### 3.1 Poisson Processes

**Frequency:** 7/7 exams (100%)    **Weight:** ~40% of exam    **Slides:** 16–19

#### 3.1.1 Key Concepts

**1. Basic Poisson Definition** If arrivals happen at rate  $\lambda$  (per unit time):

- Number of arrivals  $N(t) \sim \text{Poisson}(\lambda t)$
- $P(N(t) = k) = \frac{(\lambda t)^k e^{-\lambda t}}{k!}$
- $E[N(t)] = \lambda t, \quad \text{Var}(N(t)) = \lambda t$

**2. Thinning Property** (APPEARS IN 6/7 EXAMS)

Thinning - Most Important Concept

If customers arrive at rate  $\lambda$  and each customer is “type A” with probability  $p$  (independently):

Type A arrivals form Poisson process with rate  $\lambda p$

**Example:** Customers arrive at rate 2/min, 40% order coffee

$\Rightarrow$  Coffee orders are  $\text{Poisson}(2 \times 0.4 = 0.8/\text{min})$

### 3. Superposition/Merging Multiple independent Poisson processes combine:

If  $N_1(t) \sim \text{Poisson}(\lambda_1 t)$ ,  $N_2(t) \sim \text{Poisson}(\lambda_2 t)$  independent

$$\Rightarrow N_1(t) + N_2(t) \sim \text{Poisson}((\lambda_1 + \lambda_2)t)$$

### 4. Competing Processes (APPEARS IN 5/7 EXAMS)

Process A at rate  $\lambda_A$ , Process B at rate  $\lambda_B$  (independent):

$$P(\text{A happens first}) = \frac{\lambda_A}{\lambda_A + \lambda_B}$$

$$E[\text{time until first event}] = \frac{1}{\lambda_A + \lambda_B}$$

### 5. Conditional Expectations (APPEARS IN 5/7 EXAMS)

#### Conditional Expectation with Poisson

Given that exactly  $n$  events occurred:

$$E[\text{Time} \mid N(T) = n] = \frac{n}{\lambda}$$

**Example:** Given 10 customers arrived at rate 2/min:

$$E[T \mid N(T) = 10] = \frac{10}{2} = 5 \text{ minutes}$$

#### 3.1.2 Example Question Types

- **F2025 Q3:** Coffee shop with multiple product types (thinning + conditional expectation)
- **S2025 Q3:** Taxi arrivals vs passenger arrivals (competing processes)
- **2022F Q1:** Turkey arrivals with weight thresholds (thinning with exponential)
- **2022S Q1:** Chicken crossing road (conditional expectation)
- **2019 Q1:** Plagues arriving (thinning with acceptance probability)

## 3.2 Random Walks & First-Step Analysis

**Frequency:** 7/7 exams (100%)    **Weight:** ~35% of exam    **Slides:** 20–22

### 3.2.1 Key Concepts

#### Random Walk Setup

- Start at position  $S_0$  (usually between barriers  $a$  and  $b$ )
- Each step: move  $+1$ ,  $-1$ , or stay, with given probabilities
- Continue until hitting  $a$  or  $b$  (boundaries)
- Hitting time:  $T = \min\{n \geq 0 : S_n = a \text{ or } S_n = b\}$

#### First-Step Analysis - Hitting Probability

Define:  $p(i) = P(\text{hit } b \text{ before } a \mid \text{start at } i)$

**Recursive formula:**

$$p(i) = p_{\text{up}} \cdot p(i+1) + p_{\text{down}} \cdot p(i-1)$$

**Boundary conditions:**

$$p(a) = 0, \quad p(b) = 1$$

**Solve the system of equations for all intermediate**

#### First-Step Analysis for HITTING PROBABILITY

#### First-Step Analysis - Expected Time

Define:  $\mu(i) = E[\text{time to hit boundary} \mid \text{start at } i]$

**Recursive formula:**

$$\mu(i) = 1 + p_{\text{up}} \cdot \mu(i+1) + p_{\text{down}} \cdot \mu(i-1)$$

**DON'T FORGET THE "+1"!** (Most common mistake)

**Boundary conditions:**

$$\mu(a) = 0, \quad \mu(b) = 0$$

#### First-Step Analysis for EXPECTED TIME

**Special Case: Symmetric Random Walk** When  $P(+1) = P(-1) = 1/2$ :

$$p(i) = \frac{i-a}{b-a}$$

$$\mu(i) = (i-a)(b-i)$$

#### Biased Random Walk (APPEARS IN 4/7 EXAMS)

When  $P(+1) \neq P(-1)$ :

- Use first-step analysis and solve system of equations
- OR use characteristic equation:  $r = pr + q/r$  where  $p = P(+1)$ ,  $q = P(-1)$
- Solve:  $pr^2 - r + q = 0 \Rightarrow r = \frac{q}{p}$  (if  $p \neq q$ )

### 3.2.2 Problem-Solving Steps

1. Draw a diagram with states  $a, i, b$
2. Identify what you're solving for:  $p(i)$  or  $\mu(i)$
3. Write the recursive formula (don't forget "+1" for  $\mu$ !)
4. Write boundary conditions
5. Solve the system of equations (usually 2–3 equations)

### 3.2.3 Example Question Types

- **F2025 Q2:** Gambler with laziness parameter (modified random walk)
- **S2025 Q1:** Flight price movement (biased walk with first-step analysis)
- **2022F Q3:** Indigestion problem (hitting time analysis)
- **2022S Q2:** Easter bunny with  $\pm 1, \pm 2$  jumps (modified random walk)
- **2021 Q3:** Time warp dance (asymmetric steps)
- **2019 Q3:** Israelites in desert (biased walk with drift)
- **2017 Q3:** Asymmetric casino game (biased walk with  $+1/-2$ )

### 3.3 Continuous Random Variables: Joint & Conditional Densities

**Frequency:** 6/7 exams (86%)    **Weight:** ~25% of exam    **Slides:** 11–13

#### 3.3.1 Key Concepts

##### 1. Joint PDF $f_{X,Y}(x, y)$

- Probability density for two continuous random variables
- Must integrate to 1:  $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1$
- **Always specify the range!** (e.g., “for  $0 < y < x < \infty$ ”)

##### 2. Marginal PDF from Joint

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy \quad (\text{integrate over } y)$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx \quad (\text{integrate over } x)$$

##### 3. Conditional PDF

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x, y)}{f_X(x)}$$

##### 4. Building Joint PDF from Conditional

$$f_{X,Y}(x, y) = f_{X|Y}(x|y) \cdot f_Y(y)$$

$$f_{X,Y}(x, y) = f_{Y|X}(y|x) \cdot f_X(x)$$

##### 5. Law of Total Expectation (SUPER IMPORTANT)

Law of Total Expectation (Tower Property)

$$E[Y] = E[E[Y|X]]$$

In integral form:

$$E[Y] = \int_{-\infty}^{\infty} E[Y|X = x] \cdot f_X(x) dx$$

**Incredibly useful for computing expectations!**

##### 6. Common Transformation: $Y = UX$ When $U \sim \text{Uniform}[0, 1]$ independent of $X$ :

- Given  $X = x$ ,  $Y$  is uniform on  $[0, x]$
- $f_{Y|X}(y|x) = \frac{1}{x}$  for  $0 < y < x$
- $f_{X,Y}(x, y) = f_{Y|X}(y|x) \cdot f_X(x) = \frac{1}{x} f_X(x)$  for  $0 < y < x$

### 3.3.2 Problem-Solving Strategy

1. Identify what's given (conditional or marginal?)
2. Build joint PDF using  $f_{X,Y}(x,y) = f_{X|Y}(x|y) \cdot f_Y(y)$  or  $f_{Y|X}(y|x) \cdot f_X(x)$
3. **Check the range carefully!** Draw a picture if needed
4. For expectations: use law of total expectation  $E[Y] = E[E[Y|X]]$

### 3.3.3 Example Question Types

- **F2025 Q1:** Insurance payments  $Y = UX$  (joint density, conditional probability)
- **S2025 Q2:** Flight delay with weather index (joint density, conditional probabilities)
- **2022F Q2:** Pie eaten by dog (conditional expectation with gamma distribution)

## 4 High Priority Topics

### HIGH PRIORITY

Learn these topics after mastering the Critical Priority topics. They appear in 40–80% of exams.

#### 4.1 Exponential Distribution & Lifetimes

**Frequency:** 5/7 exams (71%)    **Weight:** ~15% of exam    **Slides:** 11, 14–15

##### 4.1.1 Exponential Distribution Basics

If  $T \sim \text{Exp}(\lambda)$ :

$$\text{PDF: } f(t) = \lambda e^{-\lambda t} \text{ for } t > 0$$

$$\text{CDF: } F(t) = 1 - e^{-\lambda t}$$

$$\text{Survival: } P(T > t) = e^{-\lambda t}$$

$$\text{Mean: } E[T] = \frac{1}{\lambda}$$

$$\text{Variance: } \text{Var}(T) = \frac{1}{\lambda^2}$$

##### 4.1.2 Memoryless Property

Memoryless Property - KEY!

$$P(T > s + t \mid T > s) = P(T > t)$$

**Interpretation:** The distribution “doesn’t remember the past”

**Key fact:** Exponential is the ONLY continuous distribution with this property!

##### 4.1.3 Minimum of Independent Exponentials

If  $T_1 \sim \text{Exp}(\lambda_1)$  and  $T_2 \sim \text{Exp}(\lambda_2)$  are independent:

$$\min(T_1, T_2) \sim \text{Exp}(\lambda_1 + \lambda_2)$$

$$P(T_1 < T_2) = \frac{\lambda_1}{\lambda_1 + \lambda_2}$$

$$E[\min(T_1, T_2)] = \frac{1}{\lambda_1 + \lambda_2}$$

**Connection to Poisson:** This relates to competing Poisson processes!

##### 4.1.4 Maximum of Exponentials

$$E[\max(T_1, T_2)] = \frac{1}{\lambda_1} + \frac{1}{\lambda_2} - \frac{1}{\lambda_1 + \lambda_2}$$

No simple closed form for the distribution of  $\max(T_1, T_2)$ .

#### 4.1.5 Example Question Types

- **2022S Q1:** Chicken crossing with exponential wait time
- **2019 Q2:** Bread rising and proofing (max of exponentials)
- **2017 Q2:** Smoker vs non-smoker lifetimes (hazard rates)

## 4.2 Brownian Motion

**Frequency:** 3/7 exams (43%)    **Weight:** ~20% when it appears    **Slides:** 23

**Note:** Increasingly common in recent exams!

### 4.2.1 Standard Brownian Motion Definition

$B(t)$  is a *standard Brownian motion* if:

1.  $B(0) = 0$  (starts at origin)
2.  $B(t)$  has continuous paths
3.  $B(t)$  has independent increments
4.  $B(t) - B(s) \sim N(0, t - s)$  for  $t > s$

### 4.2.2 Key Properties

$$\begin{aligned} E[B(t)] &= 0 \\ \text{Var}(B(t)) &= t \\ B(t) &\sim N(0, t) \\ E[B(t)^2] &= t \quad (\text{since } \text{Var} = E[X^2] - (E[X])^2 = E[X^2]) \end{aligned}$$

### 4.2.3 Conditional Expectations

For  $t > s$ :

$$\begin{aligned} E[B(t) \mid B(s) = x] &= x \\ E[B(t)^2 \mid B(s) = x] &= x^2 + (t - s) \end{aligned}$$

### 4.2.4 Common Application: Stock Prices

Geometric Brownian motion:

$$S(t) = S_0 \cdot e^{B(t)}$$

Used extensively in financial modeling.

### 4.2.5 Example Question Types

- **2022S Q3:** Stock price  $S(t) = e^{B(t)}$  (probabilities with BM)
- **2021 Q1:** Two ants doing independent Brownian motion (distance calculation)

## 5 Key Formulas & Techniques

### 5.1 Poisson Processes

#### Poisson Process Formulas

If  $N(t) \sim \text{Poisson}(\lambda t)$ :

$$P(N(t) = k) = \frac{(\lambda t)^k e^{-\lambda t}}{k!}$$

$$E[N(t)] = \lambda t$$

$$\text{Var}(N(t)) = \lambda t$$

**Thinning:** If arrivals are rate  $\lambda$ , each type A with probability  $p$ :

$\Rightarrow$  Type A arrivals are  $\text{Poisson}(\lambda p)$

**Competing:**

$$P(\text{A before B}) = \frac{\lambda_A}{\lambda_A + \lambda_B}$$

**Conditional:**

$$E[T \mid N(T) = n] = \frac{n}{\lambda}$$

### 5.2 Random Walk First-Step Analysis

#### First-Step Analysis Formulas

**For hitting probability  $p(i)$ :**

$$p(i) = p_{\text{up}} \cdot p(i+1) + p_{\text{down}} \cdot p(i-1) + p_{\text{stay}} \cdot p(i)$$

**Boundaries:**  $p(a) = 0$ ,  $p(b) = 1$

**For expected hitting time  $\mu(i)$ :**

$$\mu(i) = 1 + p_{\text{up}} \cdot \mu(i+1) + p_{\text{down}} \cdot \mu(i-1) + p_{\text{stay}} \cdot \mu(i)$$

**Boundaries:**  $\mu(a) = 0$ ,  $\mu(b) = 0$

**For symmetric walk ( $p = q = 1/2$ ):**

$$p(i) = \frac{i - a}{b - a}$$

$$\mu(i) = (i - a)(b - i)$$

### 5.3 Exponential Distribution

#### Exponential Distribution Formulas

If  $T \sim \text{Exp}(\lambda)$ :

$$\begin{aligned}f(t) &= \lambda e^{-\lambda t} \\E[T] &= \frac{1}{\lambda} \\P(T > t) &= e^{-\lambda t}\end{aligned}$$

**Memoryless:**  $P(T > s + t \mid T > s) = P(T > t)$

**Min of exponentials:**

$$\begin{aligned}\min(T_1, T_2) &\sim \text{Exp}(\lambda_1 + \lambda_2) \\P(T_1 < T_2) &= \frac{\lambda_1}{\lambda_1 + \lambda_2}\end{aligned}$$

### 5.4 Continuous Random Variables

#### Joint & Conditional Density Formulas

**Joint to Marginal:**

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$$

**Conditional:**

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

**Law of Total Expectation:**

$$\begin{aligned}E[Y] &= E[E[Y|X]] \\&= \int_{-\infty}^{\infty} E[Y|X = x] \cdot f_X(x) dx\end{aligned}$$

## 5.5 Brownian Motion

### Brownian Motion Formulas

If  $B(t)$  is standard Brownian motion:

$$B(0) = 0$$

$$E[B(t)] = 0$$

$$\text{Var}(B(t)) = t$$

$$B(t) \sim N(0, t)$$

$$B(t) - B(s) \sim N(0, t - s) \quad \text{for } t > s$$

**Conditional:**

$$E[B(t)^2 \mid B(s) = x] = x^2 + (t - s) \quad \text{for } t > s$$

## 6 Common Mistakes to Avoid

### 6.1 Poisson Processes

- ✗ Forgetting to use thinning when events split into types
- ✗ Not recognizing when to use conditional expectation  $E[T \mid N = n]$
- ✓ Always check if arrivals split by type  $\rightarrow$  use thinning

### 6.2 Random Walks

- ✗ Forgetting the “+1” in the expected time formula:  $\mu(i) = \mathbf{1} + \dots$
- ✗ Wrong boundary conditions (should be 0 at both barriers for  $\mu$ )
- ✓ Write out the recursive formula carefully with all probabilities

### 6.3 Joint Densities

- ✗ Not checking the range/support of the distribution
- ✗ Forgetting to normalize (density must integrate to 1)
- ✓ Always specify the region where  $f(x, y) > 0$

### 6.4 Exponential

- ✗ Confusing rate  $\lambda$  with mean (mean =  $1/\lambda$ )
- ✗ Not using memoryless property when applicable
- ✓ Remember: min of exponentials  $\rightarrow$  sum of rates

## 7 Detailed Exam Breakdown

### 7.1 Fall 2025 Midterm

**Q1 (Insurance — 30 pts):** Joint/Conditional Densities, Transformations ( $Y = UX$ )

- Topics: Continuous RVs, conditional probability, tail distributions
- Slides: 11–13

**Q2 (Sleepy Gambler — 30 pts):** Modified Random Walk with laziness parameter

- Topics: First-step analysis, hitting probabilities, expected hitting times
- Slides: 20–21

**Q3 (Coffee Shop — 30 pts):** Poisson with Thinning, Conditional Expectation

- Topics: Thinning property, conditional expectations, Poisson processes
- Slides: 16–18

### 7.2 Spring 2025 Midterm

**Q1 (Flight Prices — 30 pts):** Biased Random Walk with barriers

- Topics: First-step analysis, hitting probabilities, expected times
- Slides: 20–22

**Q2 (Weather Index — 30 pts):** Joint Density, Exponential conditioning

- Topics: Joint PDFs, conditional probabilities with continuous RVs
- Slides: 11–13, 14–15

**Q3 (Taxi Stop — 30 pts):** Competing Poisson Processes

- Topics: Poisson processes, conditional expectations, competing processes
- Slides: 16–18

### 7.3 Fall 2022 Midterm

**Q1 (Turkeys — 30 pts):** Poisson with Thinning, Competing Processes

- Topics: Thinning (exponential threshold), competing Poisson vs customer arrivals
- Slides: 16–18

**Q2 (Pie — 30 pts):** Conditional Expectation, Gamma Distribution

- Topics: Law of total expectation, conditional expectations with continuous RVs
- Slides: 11–13

**Q3 (Indigestion — 30 pts):** Random Walk, Hitting Time

- Topics: Discrete random walk, probability of never hitting barrier
- Slides: 20–21

## 7.4 Spring 2022 Midterm

**Q1 (Chicken Crossing — 30 pts):** Poisson + Exponential, Conditional Expectation

- Topics: Poisson count, conditional expectation given count
- Slides: 16–18, 11

**Q2 (Easter Bunny — 30 pts):** Random Walk with jumps of  $\pm 1$ ,  $\pm 2$

- Topics: Modified random walk, first-step analysis
- Slides: 20–22

**Q3 (Stock Price — 30 pts):** Brownian Motion,  $e^{B(t)}$

- Topics: Brownian motion properties, probabilities with BM
- Slides: 23

## 7.5 2021 Midterm

**Q1 (Tiger-ants — 30 pts):** Two Independent Brownian Motions

- Topics: Conditional expectations with BM, distance between BMs
- Slides: 23

**Q2 (Fruity Yogurt — 30 pts):** Poisson with Thinning (two-stage)

- Topics: Thinning with multiple probabilities, conditional expectations
- Slides: 16–18

**Q3 (Time Warp — 30 pts):** Modified Random Walk

- Topics: First-step analysis with asymmetric steps
- Slides: 20–22

## 7.6 2019 Midterm

**Q1 (Plagues — 30 pts):** Poisson with Thinning/Acceptance

- Topics: Thinning with acceptance probability, types of events
- Slides: 16–18

**Q2 (Matzot — 30 pts):** Max of Two Exponentials

- Topics: Expected max, probability calculations with exponentials
- Slides: 11, 14–15

**Q3 (Desert — 30 pts):** Random Walk with Drift

- Topics: Biased random walk, first-step analysis
- Slides: 20–22

## 7.7 2017 Midterm

### Q1 (Twitter — 30 pts): Multiple Competing Poisson Processes

- Topics: Superposition, competing processes, conditional expectations
- Slides: 16–18

### Q2 (Smoking — 30 pts): Hazard Functions, Lifetimes

- Topics: Hazard rate, probability one outlasts another
- Slides: 14–15

### Q3 (Asymmetric Game — 30 pts): Biased Random Walk ( $+1/-2$ )

- Topics: First-step analysis with asymmetric steps
- Slides: 20–22

## 8 Study Strategy & Learning Roadmap

### 8.1 Suggested 4-Day Study Plan

#### 8.1.1 Day 1: Poisson Processes (6–8 hours)

##### Morning (3–4 hours):

- Read Slides 16–19 carefully
- Take notes on thinning, superposition, competing processes
- Work through slide examples

##### Afternoon (3–4 hours):

- Practice F2025 Q3 (Coffee shop)
- Practice S2025 Q3 (Taxi stop)
- Practice 2019 Q1 (Plagues)
- Check solutions, understand every step

##### Evening Review:

- Can you explain thinning without looking at notes?
- Can you set up competing process problems?

#### 8.1.2 Day 2: Random Walks (6–8 hours)

##### Morning (3–4 hours):

- Read Slides 20–22 carefully
- Focus on first-step analysis technique
- Write out formula templates

##### Afternoon (3–4 hours):

- Practice F2025 Q2 (Sleepy gambler)
- Practice S2025 Q1 (Flight prices)
- Practice 2017 Q3 (Asymmetric game)

##### Evening Review:

- Write first-step formulas from memory
- Check: do you remember the “+1” for expected time?

### 8.1.3 Day 3: Joint Densities + Mixed Practice (6–8 hours)

#### Morning (2–3 hours):

- Read Slides 11–13
- Focus on conditional densities and law of total expectation
- Work through examples

#### Afternoon (3–4 hours):

- Practice F2025 Q1 (Insurance)
- Practice S2025 Q2 (Weather index)
- Do a FULL timed practice exam (F2025 or S2025)

#### Evening (1–2 hours):

- Quick review of Exponential (Slides 11, 14–15)
- If time: Brownian Motion basics (Slide 23)

### 8.1.4 Day 4: Final Review + Weak Areas (4–6 hours)

#### Morning (2–3 hours):

- Do another full timed practice exam
- Identify weak topics

#### Afternoon (2–3 hours):

- Focus on problems you got wrong
- Review key formulas
- Create/refine your cheat sheet

#### Evening:

- Light review, no new material
- Get good sleep before exam!

## 8.2 Emergency 2-Day Crash Plan

If you only have 2 days:

### 8.2.1 Day 1: Poisson + Random Walks ONLY (10–12 hours)

- These are 100% guaranteed to appear
- Master thinning and first-step analysis
- Do F2025 Q2, Q3 and S2025 Q1, Q3

### 8.2.2 Day 2: Joint Densities + Full Practice (10–12 hours)

- Learn conditional density basics
- Do 2 full timed practice exams
- Focus on understanding solutions

**Note:** Skip Brownian Motion if pressed for time. Focus on the big 3!

## 8.3 How to Learn Each Topic Effectively

### 8.3.1 1. Read Slides First

- Don't rush, understand each example
- Write down key formulas in your own notes
- Try to explain concepts in your own words

### 8.3.2 2. Practice Problems Immediately

- Don't just read solutions!
- Attempt problem yourself first (even if you struggle)
- Then check solution and understand each step

### 8.3.3 3. Pattern Recognition

- Notice: "When I see X, I should use technique Y"
- Example: "When I see 'types of customers' → Thinning!"
- Example: "When I see 'hitting probability' → First-step analysis!"

### 8.3.4 4. Active Recall

- After learning a topic, close your notes
- Try to write key formulas from memory
- Explain the technique out loud to yourself

### 8.3.5 5. Interleaved Practice

- Don't do 10 Poisson problems in a row
- Mix topics: Poisson → Random Walk → Joint Density
- Helps you recognize which technique to use

## 9 Quick Reference: Which Technique When?

Table 2: Problem Type Recognition Guide

| When you see...                               | Use this technique                                            |
|-----------------------------------------------|---------------------------------------------------------------|
| “Poisson process” + “types/categories”        | <b>Thinning</b>                                               |
| “random walk” + “probability of hitting”      | <b>First-Step Analysis (hitting prob)</b>                     |
| “random walk” + “expected time”               | <b>First-Step Analysis (expected time)</b>                    |
| “given that $N(t) = n$ ”                      | <b>Conditional Expectation</b> with Poisson                   |
| “exponential” + “minimum” or “first to occur” | <b>Min of Exponentials</b>                                    |
| “joint density” or “ $f(x, y)$ ”              | Set up <b>Joint PDF</b> , check range                         |
| “given $X = x$ , find $E[Y]$ ”                | <b>Conditional Expectation/Conditional Density</b>            |
| “Brownian motion” or “ $B(t)$ ”               | Use <b>BM properties</b> ( $E[B(t)] = 0$ , $\text{Var} = t$ ) |

## 10 Practice Problem Recommendations

### 10.1 Must-Do Problems (Do these first!)

1. **F2025 Q3** — Poisson thinning + conditional expectation
2. **F2025 Q2** — Random walk with modified probabilities
3. **S2025 Q1** — Biased random walk first-step analysis
4. **S2025 Q3** — Competing Poisson processes
5. **2022F Q1** — Poisson with exponential threshold
6. **2022S Q1** — Poisson + exponential conditional

### 10.2 Good Practice Problems

7. **F2025 Q1** — Joint density  $Y = UX$
8. **S2025 Q2** — Joint density with exponentials
9. **2021 Q2** — Poisson two-stage thinning
10. **2019 Q3** — Biased random walk

### 10.3 Advanced/Optional

11. **2022S Q3** — Brownian motion (stock prices)
12. **2021 Q1** — Two Brownian motions
13. **2017 Q2** — Hazard functions

## 11 Exam Day Tips

1. **Read the problem carefully** — identify which technique to use
2. **Write down what you know** — given distributions, parameters, etc.
3. **Set up the problem** — define random variables clearly
4. **Show your work** — partial credit is awarded!
5. **Check boundary conditions** — especially for random walks
6. **Simplify your answer** — reduce fractions, combine terms
7. **Sanity check** — does your answer make intuitive sense?

## 12 Knowledge Check: Are You Ready?

Before the exam, you should be able to:

### 12.1 Poisson Processes

- ☐ Explain what thinning means and when to use it
- ☐ Set up  $P(\text{A before B})$  for competing processes
- ☐ Use  $E[T \mid N = n]$  correctly

### 12.2 Random Walks

- ☐ Write first-step analysis formula from memory
- ☐ Remember to include “+1” for expected time formula
- ☐ Identify and write correct boundary conditions
- ☐ Solve a system of 2–3 linear equations

### 12.3 Joint/Conditional Densities

- ☐ Build joint PDF from conditional + marginal
- ☐ Specify correct range for density
- ☐ Use law of total expectation  $E[Y] = E[E[Y|X]]$
- ☐ Find marginal from joint PDF

### 12.4 Exponential

- ☐ Know memoryless property
- ☐ Set up min of exponentials
- ☐ Remember  $E[T] = 1/\lambda$  (not  $\lambda!$ )

## 13 Final Motivation

### Key Insight

#### Remember:

- You have a clear roadmap now!
- Topics are well-defined and learnable
- Focus on understanding, not memorizing
- Past students have done this successfully
- You can learn a lot in a focused few days

### 13.1 Strategy for Success

1. Don't try to learn everything perfectly
2. Focus on high-frequency topics first
3. Understand the **TECHNIQUES**, not just formulas
4. Practice with actual exam problems
5. Learn from mistakes

### 13.2 You've Got This!

The fact that you're being proactive and strategic shows you have what it takes to succeed.

**Now stop reading and start learning!**

Good luck! You can do this!